4.2 Data Flow Design

The data flow begins with receiver frames and ends with browser visualization and export files. The decoder validates supported frame layouts and merges partial updates into an ICAO-keyed aircraft state. The preprocessor normalizes available fields, while the storage layer excludes records whose latitude and longitude are both zero and suppresses unnecessary duplicate trajectory samples. This preserves useful partial identity or motion information without treating an unusable position as a valid track point.

Figure 2: Level-1 Data Flow Diagram

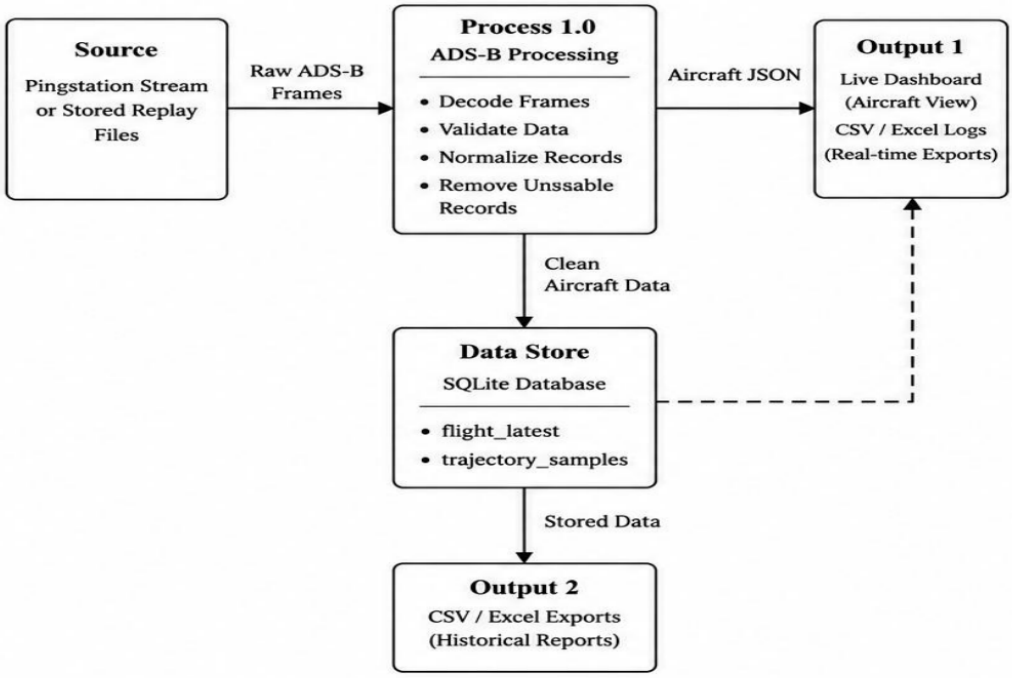


Figure 2: Level-1 Data Flow

4.3 Database Design

The SQLite database has two central tables. The flight_latest table stores one current record per flight, while trajectory_samples stores time-series samples for movement history. This design keeps live lookup fast and still preserves historical data for trajectory analysis.